

Sums of Squares and Orthogonal Decomposition

An analysis of variance table looks like bookkeeping. Behind every row there is a subspace, and behind the whole table an orthogonal decomposition of the space in which the data live: sums of squares are squared lengths of projections, and degrees of freedom are dimensions. Sequential and partial sums of squares, and the Type I, II and III sums of squares printed by software, differ only in which subspaces they project onto, and so in which hypotheses they address. When the subspaces are orthogonal, as in balanced designs and with orthogonal polynomials, the distinctions disappear. The coefficient of determination is a ratio of two of these squared lengths, and we mark what it does not measure. Finally, the expectation of each sum of squares shows what an F ratio compares, preparing the tests of Part III.

What you need. [Chapter 6](#) throughout, especially projections and their ranks (Theorem 6.3.7, Proposition 6.3.4), nested models (Theorem 6.5.1, Theorem 6.5.2), the Frisch–Waugh–Lovell theorem (Theorem 6.6.2) and R^2 as a squared cosine (Theorem 6.7.2). The linear model and the moments of least squares estimates from [Chapter 5](#) (Definition 5.3.1, Theorem 5.5.1). The mean of a quadratic form (Theorem 2.3.1) and the distribution theory of [Chapter 4](#) (Theorem 4.4.4, Theorem 4.5.1). Estimable functions and contrasts from [Chapter 8](#) appear in [Sections 9.2 to 9.4](#). Three facts from [Chapter 1](#) are used once each: rank and nullity and the partitioned inverse in [Section 9.2](#) (Theorem 1.2.2, Theorem 1.6.4), and the Cauchy–Schwarz inequality in [Section 9.5](#) (Proposition 1.1.1).

9.-1.1 Roadmap

- [Decomposing the total sum of squares](#)
- [Extra sums of squares](#)
- [Sequential and partial sums of squares](#)
- [Orthogonal designs](#)
- [The coefficient of determination and its limits](#)
- [Expected mean squares](#)
- [Summary and notes](#)